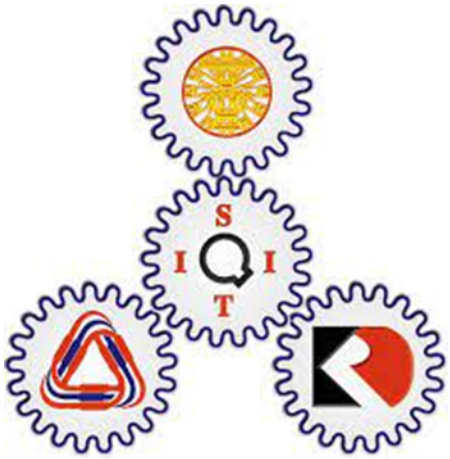
Sirindhorn International Institute of Technology



Semester 2/2023

ITS229 Human-Computer Interface Design

Group Assignment #2

Group 8: *Vraiment Cool*

Project Topic: Play N' Learn - A gamified learning experience

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Objectives

• Creating a project topic based around a problem, and conducting research to understand the context and specific needs of the user for said project topic.

• Proposing a solution, and a comprehensive list of requirements for the solution.

Introduction

This document will serve as a report file for ITS/DES229 2023-2 Group Assignment #2. In this assignment, we will propose a project topic for our product development. Then, we will conduct research into the population to gather information related to our topic. Lastly, we will perform an in-depth analysis of the information to create a set of requirements for our product.

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Roles

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Proposed Topic

Gamified Learning Experience

In the rapidly evolving landscape of modern education, the traditional models of learning are facing a myriad of challenges. From disengagement and lack of motivation to the struggle of adapting to diverse learning styles, educational institutions are seeking innovative solutions to captivate the attention and enthusiasm of students. One promising avenue in this quest for educational transformation is the integration of gamified experiences into the learning process.

Gamification is a dynamic approach that brings elements of games into non-game contexts, such as education, to enhance engagement and motivation. In the realm of education, gamification involves integrating game-like features, such as points, levels, rewards, and challenges, into the learning experience. This strategy taps into the innate human desire for competition, achievement, and a sense of progress.

By turning educational content into a more interactive and rewarding format, gamification helps make learning enjoyable and captivating for students. It encourages active participation, critical thinking, and problem-solving, fostering a positive and dynamic learning environment. Ultimately, gamification has the potential to transform education by making it more engaging, motivating, and aligned with the preferences and behaviors of today's learners.

In summary, **we aim to utilize the concept of gamification to create a product that will allow educators to bring a gamified learning experience to the classroom**, which will allow learners to become more engaged in the lessons, and increase their effectiveness in retaining and applying the knowledge they have obtained.

Research

Overview

In order to gather data for our project topic, we have chosen to conduct research via the following methods.

1. **Surveys:**

   This is done through Google Forms. Respondents are asked several questions regarding their opinions on the teaching methods in modern education, and their ideas on how it could be improved.

2. **Relevant Studies and Research Papers:**

   Studies relevant to the topic of effectiveness in different teaching methods are included to support our analysis and conclusions.

3. **Researching Similar Products:**

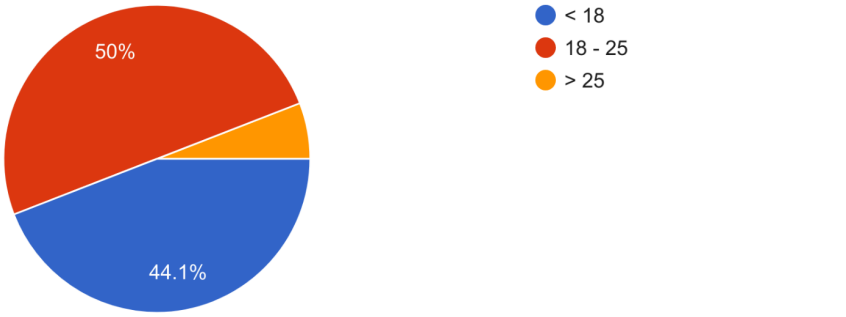
   Products with similar ideas and functions are researched in order to help us establish requirements that better reflect the actual needs of users.

Survey

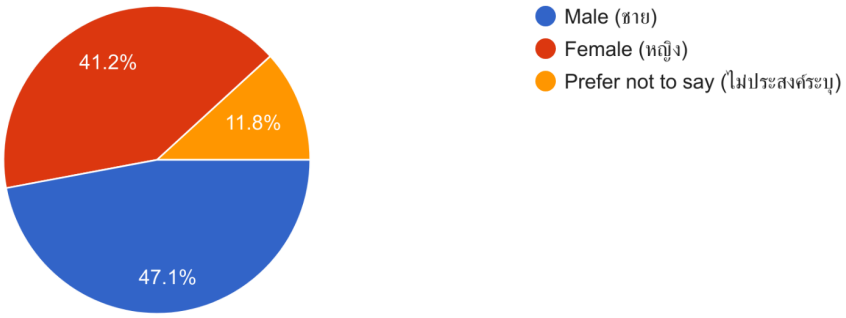
The survey is conducted through Google Forms, which contains 12 questions related to the user's opinions on the teaching methods in modern education, and their ideas on how it could be improved. **34 responses** were collected throughout the survey period.

The survey is separated into two sections; Demographics (4 questions), and Opinions (8 questions).

Age - อายุ
34 responses

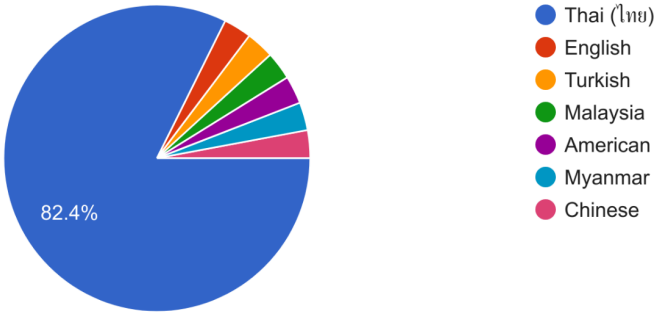


Gender - เพศ
34 responses



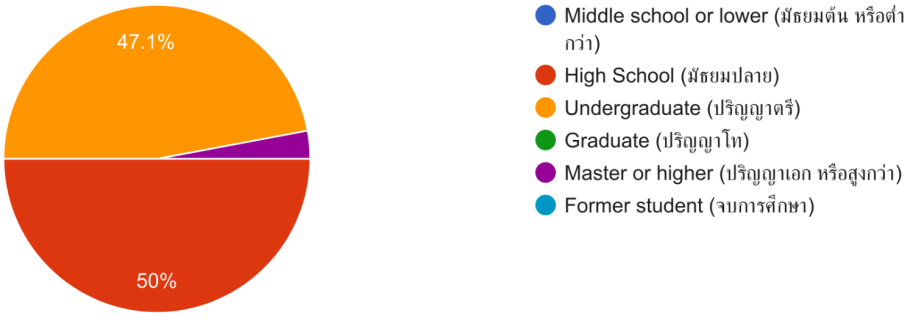
Nationality - สัญชาติ

34 responses



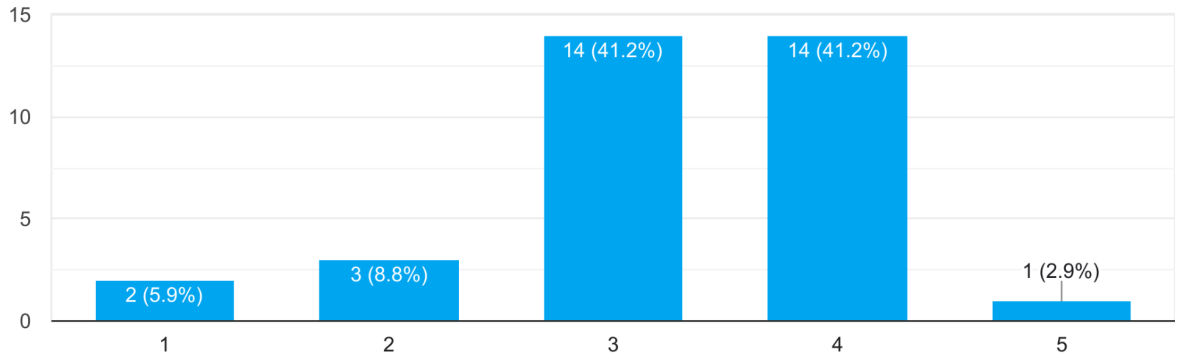
Degree of Education - ระดับการศึกษา

34 responses

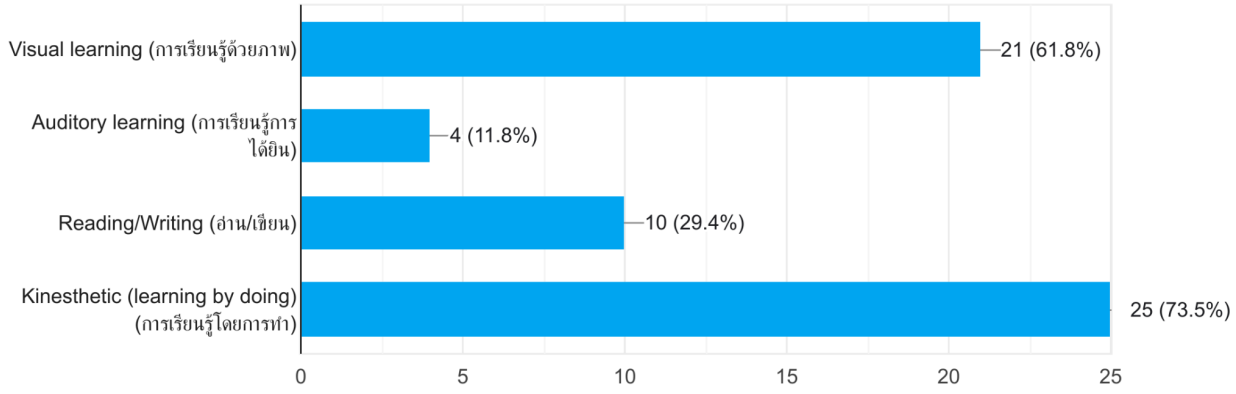


From the Demographics section, we observe that the majority of respondents are Thai male undergraduate or highschool students.

How satisfied are you with the educational style used by your teachers and instructors ? -
คุณรู้สึกพึงพอใจกับวิธีการสอนของคุณครูและอาจารย์เพียงใด
34 responses

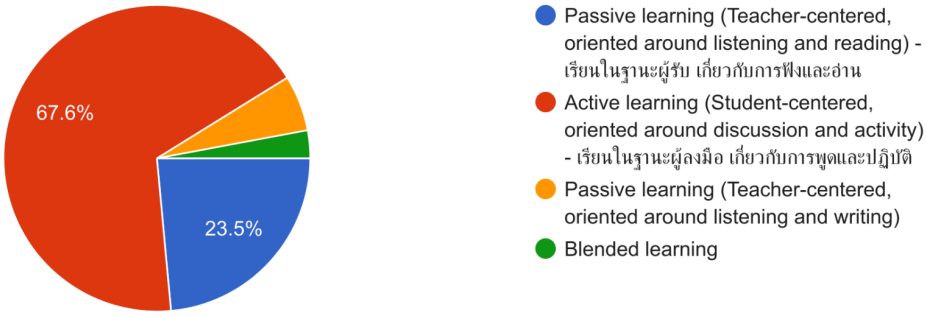


Which learning style suits you the best? (2 maximum) -
รูปแบบการเรียนรู้แบบใดที่เหมาะสมกับคุณ/ที่คุณชอบที่สุด (อย่างมาก 2 ข้อ)
34 responses



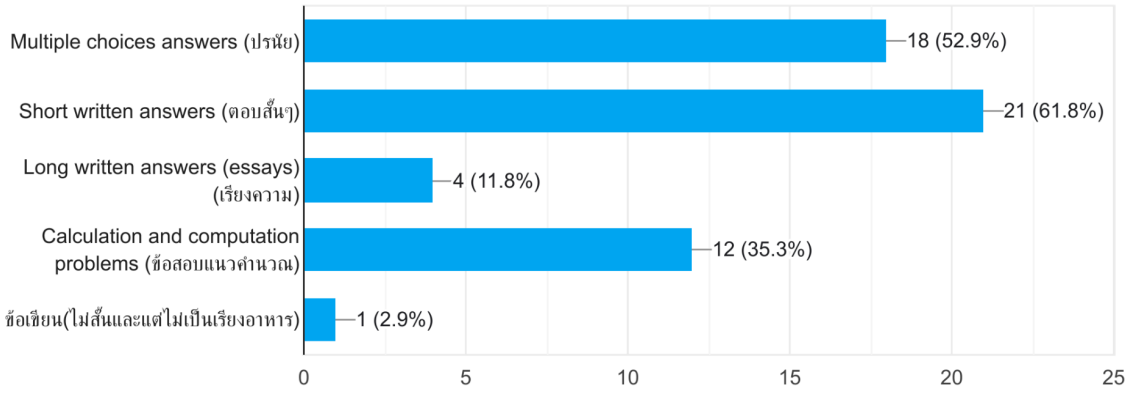
Which of the following teaching method is more effective/preferable for you? - รูปแบบการสอนแบบใดที่เหมาะสมกับคุณ/ที่คุณชอบที่สุด

34 responses

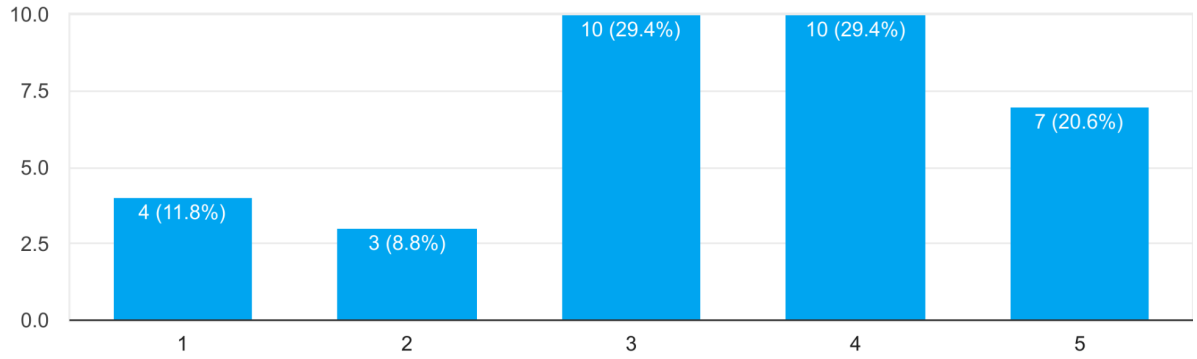


Which evaluation style do you prefer? (2 maximum) - รูปแบบการทดสอบความรู้แบบใดที่เหมาะสมกับคุณ/ที่คุณชอบที่สุด (อย่างมาก 2 ข้อ)

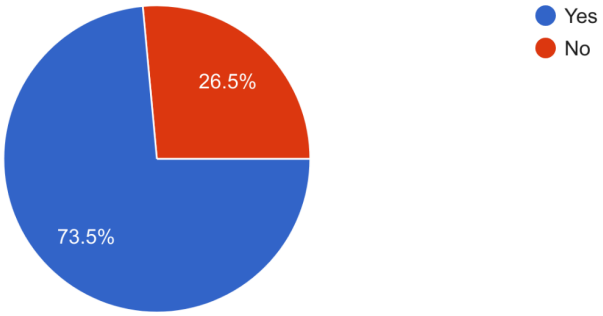
34 responses



How much do you tend to compare your academic results to others? -
คุณมีแนวโน้มชอบเปรียบเทียบผลการเรียนของคุณกับคนอื่นๆ มากเพียงใด
34 responses



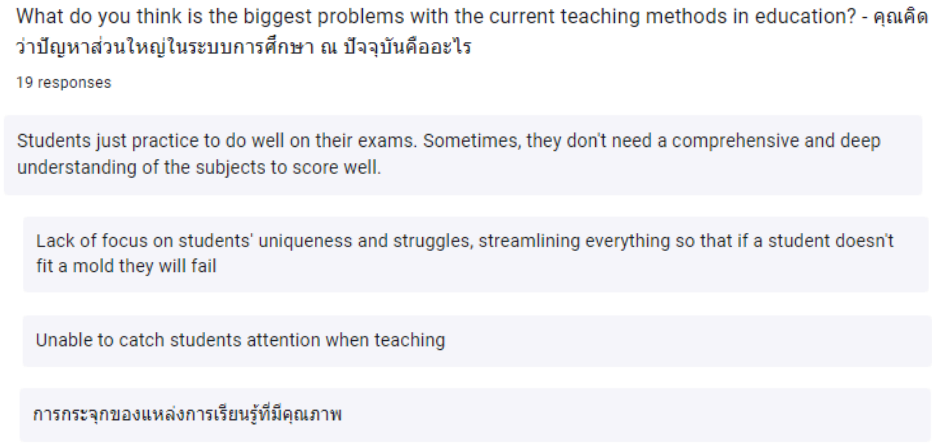
Have you ever played games that taught you something? (e.g. quizzes, puzzles, video games) -
คุณเคยเล่นเกมที่ช่วนให้คุณได้เรียนรู้หรือไม่
34 responses



From the Opinions section, we observe that the majority of respondents:

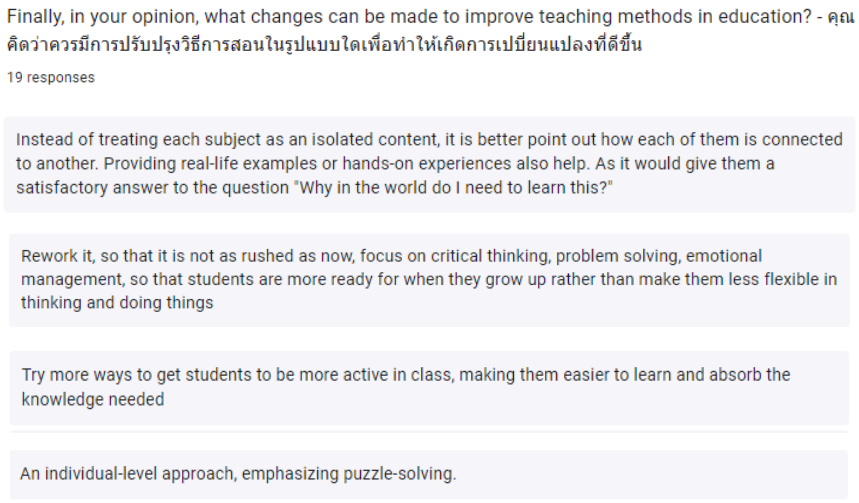
1. are somewhat satisfied with the teaching style used by their teachers and instructors.
2. feel that visual learning and kinesthetic (learning by doing) suit them the most.
3. feel that active learning is more effective/preferable for them.
4. prefer multiple-choice questions and short-written answers for knowledge evaluation.
5. tend to compare their academic performance results to those of other students.
6. have played games that taught them something.

Additionally, for the last two long-answer opinion questions, certain key points can be summarized as follows:



The biggest problems with current teaching methods:

1. The current teaching methods value rote memorization over comprehension.
2. Class schedules are overly long & lacking in engaging activities.
3. Too focused on streamlining the lessons & disregarding different capabilities of each student, causing some people to fall behind despite trying to the best of their abilities.
4. Great disparity in the quality of education for different areas.



Changes that can be made to improve teaching methods:

1. Emphasizing the interconnectedness between different branches of knowledge, particularly in real-world, job-related applications.
2. Foster the comprehension of core principles instead of rote memorization, and allow students to develop their intuition via action.
3. More focus on the individual's capabilities.
4. Apply more active learning strategies to engage the students.

Research

Our short time survey highlights our observations regarding the problem with current teaching methods in academic schools. In this section, we will showcase relevant research and studies to strongly back-up those opinions hence inducing the need of our solution. References will all be put in the reference section.

1) Where's the evidence that active learning works ?(12/2006)

This article was released in 2006 on the American Physiological Society 's website. The American Physiological Society is a non-profit professional society for physiologists. It has nearly 10,000 members, most of whom hold doctoral degrees in medicine, physiology.
This article aimed to regroups all of the existing proofs of active learning benefits, as back in 2006 more and more studies were published promoting the benefits of active learning over the classical academic model. Different studies and experiments were involved and carefully analyzed in this study, on these following facts :

- Evidence From the Learning Sciences, Cognitive Science, and Educational Psychology Supporting Active Learning
- Some things that are learned are specific to the domain or context (subject matter or course) in which they were learned, whereas other things are more readily transferred to other domains.
- Individuals are likely to learn more when they learn with others than when they learn alone.
- Meaningful learning is facilitated by articulating explanations, whether to one's self, peers, or teachers.

After defining the different domain of studies as well as the difference between active and passive learning, the study concluded:

"*There IS evidence that active learning, student-centered approaches to teaching physiology work, and they work better than more passive approaches. There is no single definitive experiment to prove this, nor can there be given the nature of the phenomena at work, but the very multiplicity of sources of evidence makes the argument compelling.*"

This study was made in 2006. Nowadays, we DO have concrete experiences made on this topic, like our two following chosen studies.

2) Can digital games in school improve attention ? (07/2018)

This study was released in 2018 in the Journal of Computers in Education, a journal that aims at making an impact on educational practices and thus to transform learning.

It was conducted on Brazilian elementary school students: Thirty students played digital games daily (15 min) at the beginning of class for 6 weeks, while a group of 41 students had the normal school routine. Their attention was measured for both groups before and after the training.

The study showed that the period (time between beginning and end of study) had a significant impact, improving the attention of students, but the “trained with games” group actually did outperform the control group. This experiment allowed the study to conclude :

“*These findings suggest that the use of digital games in school routine can enhance the cognitive improvement that is already obtained in the normal school routine, creating an enriched environment to stimulate students’ cognitive development.*”.

This conclusion aligns with our observations made from our survey in the first section.

3) Case study : The added value of a gaming context and intelligent adaptation for a mobile learning application for vocabulary learning (07/2014)

This last study was also conducted in the “Computers & Education” journal, a journal aiming to transform learning and educational practices using new technologies, in the 76th volume in July 2014.

Two groups were picked with different teaching application in order to teach english vocabulary during two weeks, the first one with a standard application “MEL-original”, and the second one with a more gamify and enhanced version “MEL-enhanced”, a copy of MEL-original’s app but embedded in an adventure game making it a whole game as is.

- Two learning themes were used: Zoo animals and Neighbourhood.
- Both groups attended lessons at school on Zoo Animals and Neighbourhood during the same periods they were allowed to work with the application at home.
- A pre- and post-test were conducted to establish the initial vocabulary knowledge and the knowledge acquired during the learning phase.

Results :

Even though both groups spent the same time learning, the “MEL-Enhanced” experimental group outperformed the “MEL-original” control group, and the the students in the experimental group valued “MEL-enhanced” more than the children from the control group valued “MEL-original”. We can conclude from this study that the need for a gamified experience is indeed beneficial and more efficient for the student.

Researching Similar Products

In order to get a better picture of the requirements of our target demographic, we have collected some information on products similar to ours.

Duolingo

Duolingo is a popular language-learning platform that offers courses in a wide variety of languages. It was developed by Luis von Ahn and Severin Hacker and was launched in 2011. Duolingo uses a gamified approach to language learning, making it engaging and interactive for users. Key features of the application include:

Gamification: Duolingo incorporates game-like elements into its language-learning experience, making it more engaging and enjoyable. Users can earn points, compete with friends, and unlock achievements, which adds a sense of accomplishment and motivation.

Interactive Learning: Duolingo employs a variety of interactive exercises, including listening, speaking, and translation tasks. This dynamic approach helps users develop different language skills, and the instant feedback provided after each task allows for continuous improvement.

Progress Tracking: Duolingo keeps track of users' progress, allowing them to see how far they've come in their language-learning journey. This sense of achievement can be motivating and encourages users to stick with their language-learning goals.

Community and Social Features: Duolingo enables users to connect with friends, compete in leagues, and share achievements. The social aspect adds a collaborative and competitive element to the learning process, fostering a sense of community among users.

Brilliant

Brilliant is an online learning platform that offers interactive courses in various fields, with a focus on mathematics, science, and engineering. The platform is designed to provide users a hands-on and active learning experience. Brilliant's courses often involve solving problems and completing interactive exercises, allowing learners to actively engage with the material. Key features include:

Interactive Learning: Brilliant emphasizes problem-solving and interactive learning through its platform. Users are presented with challenging problems and exercises that encourage critical thinking and application of concepts.

Courses in Various Subjects: *While Brilliant is known for its strong emphasis on mathematics and science, it also offers courses in other subjects such as computer science, data science, physics, and more.*

Progress Tracking: *Users can track their progress through the courses, and the platform provides immediate feedback on problem-solving attempts. This real-time feedback helps learners understand concepts more deeply and correct misunderstandings promptly.*

Daily Challenges: *Brilliant offers daily challenges to keep users engaged and to reinforce their learning on a consistent basis. These challenges are designed to be short and accessible, encouraging regular practice.*

Practical Application: *Courses on Brilliant often emphasize the practical application of concepts. Users are encouraged to apply what they've learned to real-world problems, promoting a more holistic understanding of the subject matter.*

Solution and Requirements

A gamified learning experience app

Our product is an app to help users create a “game-ified” learning experience. Borrowing concepts from flashcard-type learning games (e.g. World Geography, Duolingo), we intend to provide a platform where users can create similar experiences with any subject they’d like. Users can either choose to create an experience (for educators), or choose to play it (for students) - with a built-in progression system similar to games. (e.g. leveling up, progression paths)

From our research, we can conclude several key ideas that will be relevant to the requirements of our product as follows:

1. Users prefer learning via visuals and action, therefore a system that allows visual interactions with the subject materials would be ideal.
2. Users mostly prefer the individualized learning style - including self-paced learning, individual-based evaluation, and self-motivation.
3. Users seem to prefer the sense of competition among peers as a motivation for learning.
4. Users prefer a back-and-forth approach to teaching with instant feedback, as opposed to a one-way approach typically found in lecture-based learning.

Functional Requirements:

- **User Authentication:** Implement a secure user authentication system to allow users to create accounts and log in between different platforms.

- **Profile Creation:** Enable users to create profiles with relevant information, such as personal preferences, subjects of interest, and educational background.

- **Creation Mode:** Develop a user-friendly interface for educators to create learning experiences. Include options for selecting subjects, adding questions, and defining progression paths. Allow the addition of multimedia elements (images, videos) to enhance the learning experience.

- **Gaming Elements:** Integrate game-like features such as leveling up, earning points, and unlocking achievements to motivate and engage learners. Implement a built-in progression system to track and display users' advancement through the learning experiences.

- **User Interaction:** Implement features for users to provide feedback on the learning experiences. Include a commenting system for discussions and clarification between educators and students.

- **Progress Tracking:** Develop a comprehensive tracking system to monitor and display the progress of students within each learning experience. Provide educators with analytics to assess the effectiveness of their created content from the history of score or time usage of playing in each different subject and question.

Non-functional Requirements:

- **Performance:** Ensure fast loading times and responsiveness within the app. Optimize the app to handle a large number of concurrent users and content that is created.

- **Security:** Put strong security measures in place to keep user information safe. Incorporate user data privacy features that adhere to applicable laws.

- **Cross-Platform Compatibility:** Make sure the software works on all platforms (PC, Android, iOS). be suitable for a range of devices.

- **Reliability:** Minimize downtime by implementing reliable hosting and server solutions. Implement automated backup mechanisms to prevent data loss.

- **Usability:** Design an intuitive and user-friendly interface for both educators and students. Conduct usability testing to identify and address any potential usability issues.

- **Accessibility:** By adhering to accessibility guidelines and taking care of these non-functional and functional requirements, make sure that the app is usable by people of various abilities. The app needs to be well-equipped to give teachers and students a fun and successful "game-ified" learning experience.

References

- Survey responses https://docs.google.com/spreadsheets/d/1qzIyrUV5sv_hekaLcIV_8Yk_oqmtwOEeQL2j3NHpOlw/edit?usp=sharing
- https://www.duolingo.com
- https://brilliant.org
- https://www.pcmag.com/reviews/duolingo

Studies (sub references are all listed in each of the studies):

- https://journals.physiology.org/doi/full/10.1152/advan.00053.2006? Importance of active learning and proofs (12/2006)
- https://link.springer.com/article/10.1007/s40692-018-0111-3 Can digital games in school improve attention ? (07/2018)
- https://www.sciencedirect.com/science/article/abs/pii/S0360131514000633?via%3Dihub Case study : The added value of a gaming context and intelligent adaptation for a mobile learning application for vocabulary learning (07/2014)